Volume I Set Theory Reference

Contents

Set Theory

0.1 Model of Set Theory

0.2 Sets

0.2.1 Sets, Membership, and ZFC Axioms

Definition 0.2.1 (Set and Membership)**.** In axiomatic set theory, the notions of *set* and *membership* are *primitive*.

- A *set* is an object.
- *Membership* is a binary relation, denoted by $\in$, between objects.

If x is an object and A is a set, the statement $x \in A$ is read as "x is an element of A." No definition of "set" or "$\in$" is given in more basic terms. Their meaning is determined entirely by the axioms governing them.

0.2.2 Set Constructions and Operations

Definition 0.2.2 (Empty Set)**.** The unique set with no elements is called the *empty set* and is denoted $\varnothing$.

Definition 0.2.3 (Subset)**.** Let A and B be sets. We say A is a *subset* of B, written $A \subseteq B$, if every element of A is also an element of B:

$$A \subseteq B \iff \forall x\,(x \in A \to x \in B).$$

Definition 0.2.4 (Proper Subset)**.** A is a *proper subset* of B, written $A \subsetneq B$, if $A \subseteq B$ and $A \neq B$.

Definition 0.2.5 (Set Equality)**.** Two sets A and B are *equal*, written $A = B$, if they have the same elements:

$$A = B \iff \forall x\,(x \in A \leftrightarrow x \in B).$$

Equivalently, $A = B \iff (A \subseteq B \land B \subseteq A)$.

Definition 0.2.6 (Union)**.** The *union* of A and B, denoted $A \cup B$, is the set of all elements belonging to at least one of the two sets:

$$A \cup B \;=\; \{\, x \mid x \in A \lor x \in B \,\}.$$

Definition 0.2.7 (Intersection)**.** The *intersection* of A and B, denoted $A \cap B$, is the set of all elements common to both:

$$A \cap B \;=\; \{\, x \mid x \in A \land x \in B \,\}.$$

Definition 0.2.8 (Set Difference)**.** The *set difference* $A \setminus B$ is the set of elements in A but not in B:

$$A \setminus B \;=\; \{\, x \mid x \in A \land x \notin B \,\}.$$

Definition 0.2.9 (Symmetric Difference)**.** The *symmetric difference* $A \triangle B$ is the set of elements belonging to exactly one of A and B:

$$A \triangle B \;=\; (A \setminus B) \cup (B \setminus A) \;=\; \{\, x \mid (x \in A \vee x \in B) \wedge x \notin A \cap B \,\}.$$

Definition 0.2.10 (Complement)**.** Let U be a fixed universe and $A \subseteq U$. The *complement* of A relative to U, denoted A^c, is:

$$A^c \;=\; U \setminus A.$$

Definition 0.2.11 (Cartesian Product)**.** The *Cartesian product* of sets A and B is:

$$A \times B = \{\, (a, b) \mid a \in A \text{ and } b \in B \,\}.$$

Definition 0.2.12 (Power Set)**.** The *power set* of A, denoted $\mathcal{P}(A)$, is the set of all subsets of A:

$$\mathcal{P}(A) \;:=\; \{\, S \mid S \subseteq A \,\}.$$

Theorem 0.2.13 (De Morgan's Laws)**.** *Let U be a universe and $A, B \subseteq U$. Then*

$$(A \cup B)^c = A^c \cap B^c \quad \text{and} \quad (A \cap B)^c = A^c \cup B^c.$$

Go to proof.

Definition 0.2.14 (Set Duality)**.** Two set-theoretic expressions over a fixed universe U are *dual* if one is obtained from the other by simultaneously replacing $\cup \leftrightarrow \cap$ and $\varnothing \leftrightarrow U$, with complements unchanged.

Corollary 0.2.15 (Principle of Set Duality)**.** *Any identity involving $\cup$, $\cap$, $\varnothing$, and U that holds for all subsets of a universe remains valid when each operation and constant is replaced by its dual.*

0.2.3 Algebraic Laws of Set Operations

Theorem 0.2.16 (Commutativity of Union and Intersection)**.** *Let A, B be sets. Then*

$$A \cup B = B \cup A \quad \text{and} \quad A \cap B = B \cap A.$$

Go to proof.

Theorem 0.2.17 (Associativity of Union and Intersection)**.** *Let A, B, C be sets. Then*

$$(A \cup B) \cup C = A \cup (B \cup C) \quad \text{and} \quad (A \cap B) \cap C = A \cap (B \cap C).$$

Go to proof.

Theorem 0.2.18 (Distributive Laws)**.** *Let A, B, C be sets. Then*

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C),$$

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C).$$

Go to proof.

Theorem 0.2.19 (Identity and Absorption Laws)**.** *Let A, B be sets and U a universe with $A \subseteq U$. Then*

$$A \cup \varnothing = A, \qquad A \cap U = A,$$

$$A \cup (A \cap B) = A, \qquad A \cap (A \cup B) = A.$$

Go to proof.

Theorem 0.2.20 (Involution of Complement)**.** *For any $A \subseteq U$,*

$$(A^c)^c = A.$$

Go to proof.

0.3 Families

0.3.1 Indexed Families of Sets

Definition 0.3.1 (Indexed Family of Sets)**.** Let I be a set (the *index set*) and U a universe. An *indexed family of sets* is a function

$$F : I \to \mathcal{P}(U),$$

with $F(i)$ typically written A_i. The family is denoted $\{A_i\}_{i\in I}$.

Definition 0.3.2 (Indexed Union)**.** The *union* of the indexed family $\{A_i\}_{i\in I}$ is:

$$\bigcup_{i\in I} A_i \;:=\; \{\, x \mid \exists i \in I \text{ such that } x \in A_i \,\}.$$

Definition 0.3.3 (Indexed Intersection)**.** For $I \neq \varnothing$, the *intersection* of $\{A_i\}_{i\in I}$ is:

$$\bigcap_{i\in I} A_i \;:=\; \{\, x \mid \forall i \in I,\ x \in A_i \,\}.$$

Theorem 0.3.4 (De Morgan's Laws for Indexed Families)**.** *Let $\{A_i\}_{i\in I}$ be an indexed family of subsets of a universe U. Then*

$$U \setminus \bigcup_{i\in I} A_i = \bigcap_{i\in I}(U \setminus A_i).$$

If $I \neq \varnothing$, then

$$U \setminus \bigcap_{i\in I} A_i = \bigcup_{i\in I}(U \setminus A_i).$$

Go to proof.

Definition 0.3.5 (Pairwise Disjoint Family)**.** The family $\{A_i\}_{i\in I}$ is *pairwise disjoint* if

$$\forall i, j \in I,\ i \neq j \to A_i \cap A_j = \varnothing.$$

Definition 0.3.6 (Cover)**.** A collection $\mathcal{C} \subseteq \mathcal{P}(A)$ is a *cover* of A if

$$\bigcup_{C\in\mathcal{C}} C = A.$$

Arbitrary Cartesian Products

Definition 0.3.7 (Arbitrary Cartesian Product)**.** Let $\{A_i\}_{i\in I}$ be an indexed family of sets. The *Cartesian product* is:

$$\prod_{i\in I} A_i \;:=\; \Big\{ f : I \to \bigcup_{i\in I} A_i \;\Big|\; \forall i \in I,\ f(i) \in A_i \Big\}.$$

0.3.2 Covers, Subcovers, and the Finite Intersection Property

Definition 0.3.8 (Cover)**.** Let A be a set and $\mathcal{C}$ a collection of sets. $\mathcal{C}$ is a *cover* of A if every element of A belongs to at least one member of $\mathcal{C}$:

$$A \;\subseteq\; \bigcup_{C\in\mathcal{C}} C.$$

Definition 0.3.9 (Subcover)**.** $\mathcal{C}'$ is a *subcover* of $\mathcal{C}$ if $\mathcal{C}' \subseteq \mathcal{C}$ and $\mathcal{C}'$ is still a cover of A.

Definition 0.3.10 (Finite Cover)**.** A cover is *finite* if it has finitely many members. A *finite subcover* is a subcover $\mathcal{C}' \subseteq \mathcal{C}$ with $|\mathcal{C}'| < \infty$.

Definition 0.3.11 (Open Cover)**.** In a topological space (X, τ), an *open cover* of a subset $A \subseteq X$ is a cover $\mathcal{C}$ of A all of whose members are open sets: $\mathcal{C} \subseteq \tau$.

Definition 0.3.12 (Finite Intersection Property)**.** A collection $\mathcal{F}$ of sets has the *finite intersection property* (FIP) if every finite subcollection has nonempty intersection: for all finite $\mathcal{F}' \subseteq \mathcal{F}$,

$$\bigcap_{F \in \mathcal{F}'} F \neq \varnothing.$$

Proposition 0.3.13 (FIP–Cover Duality)**.** *Let X be a set, $A \subseteq X$, and $\{U_\alpha\}_{\alpha \in I}$ a family of subsets of X. Let $F_\alpha = U_\alpha^c = X \setminus U_\alpha$. Then:*

$$\{U_\alpha\} \text{ covers } A \iff \{F_\alpha \cap A\}_{\alpha \in I} \text{ does not have the FIP.}$$

More precisely: the family $\{U_\alpha\}$ does not *cover A if and only if the family of closed complements $\{F_\alpha\}$ has the FIP relative to A.* *Go to proof.*

Proofs

Relations

0.4 Relations — Quick Reference

0.4.1 Ordered Pairs and Relations

Definition 0.4.1 (Ordered Pair)**.** Let a and b be sets. The *ordered pair* (a, b) is defined (Kuratowski) as

$$(a, b) \ := \ \{\{a\}, \{a, b\}\}.$$

Theorem 0.4.2 (Uniqueness of Ordered Pairs)**.** *For any sets a, b, c, d,*

$$(a, b) = (c, d) \iff (a = c \wedge b = d).$$

Go to proof.

Definition 0.4.3 (Cartesian Product — as foundation for relations)**.** The *Cartesian product* of sets A and B is:

$$A \times B \ := \ \{\, (a, b) \mid a \in A \text{ and } b \in B \,\}.$$

Definition 0.4.4 (Relation)**.** A *relation* from A to B is any subset $R \subseteq A \times B$.
If $(a, b) \in R$ we write $a\,R\,b$ and say "a is related to b." When $A = B$ we say R is a *relation on* A.

Definition 0.4.5 (Domain of a Relation)**.** Let $R \subseteq A \times B$ be a relation. The *domain* of R is

$$\mathrm{Dom}(R) \ := \ \{\, a \in A \mid \exists b \in B,\ (a, b) \in R \,\}.$$

Definition 0.4.6 (Range of a Relation)**.** Let $R \subseteq A \times B$ be a relation. The *range* of R is

$$\mathrm{Ran}(R) \ := \ \{\, b \in B \mid \exists a \in A,\ (a, b) \in R \,\}.$$

Tarski also calls this class the *counter-domain* or *converse domain*.

Definition 0.4.7 (Converse Relation)**.** If $R \subseteq A \times B$, the *converse* of R is the relation $R^{-1} \subseteq B \times A$ defined by

$$(b, a) \in R^{-1} \iff (a, b) \in R.$$

Equivalently, $b\,R^{-1}\,a$ iff aRb.

Definition 0.4.8 (Identity Relation)**.** For a set A, the *identity relation* on A is

$$\Delta_A \ := \ \{\, (a, a) \mid a \in A \,\}.$$

Definition 0.4.9 (Composition of Relations)**.** Let $R \subseteq A \times B$ and $S \subseteq B \times C$. The *composition* $S \circ R \subseteq A \times C$ is defined by

$$(a, c) \in S \circ R \iff \exists b \in B,\ (a, b) \in R \text{ and } (b, c) \in S.$$

Theorem 0.4.10 (Associativity of Relation Composition)**.** *Let $R \subseteq A \times B$, $S \subseteq B \times C$, and $T \subseteq C \times D$. Then*

$$T \circ (S \circ R) = (T \circ S) \circ R.$$

Go to proof.

Theorem 0.4.11 (Identity Laws for Relation Composition)**.** *Let $R \subseteq A \times B$. Then*

$$R \circ \Delta_A = R \quad \textit{and} \quad \Delta_B \circ R = R.$$

Go to proof.

Definition 0.4.12 (Null Relation)**.** For sets A and B, the *null relation* from A to B is $\varnothing \subseteq A \times B$. No element of A is related to any element of B.

Definition 0.4.13 (Universal Relation)**.** For sets A and B, the *universal relation* from A to B is $A \times B$ itself. Every element of A is related to every element of B.

Definition 0.4.14 (Relation Inclusion)**.** Let $R, S \subseteq A \times B$. Relation inclusion is ordinary set inclusion:

$$R \subseteq S \iff \forall a \in A \, \forall b \in B, \; ((a, b) \in R \Rightarrow (a, b) \in S),$$

Definition 0.4.15 (Relation Equality)**.** Let $R, S \subseteq A \times B$. Relation equality is ordinary set equality:

$$R = S \iff R \subseteq S \text{ and } S \subseteq R.$$

Definition 0.4.16 (Union of Relations)**.** Let $R, S \subseteq A \times B$. The *union* $R \cup S$ is the relation with

$$(a, b) \in R \cup S \iff (a, b) \in R \text{ or } (a, b) \in S.$$

Definition 0.4.17 (Intersection of Relations)**.** Let $R, S \subseteq A \times B$. The *intersection* $R \cap S$ is the relation with

$$(a, b) \in R \cap S \iff (a, b) \in R \text{ and } (a, b) \in S.$$

Definition 0.4.18 (Difference of Relations)**.** Let $R, S \subseteq A \times B$. The *difference* $R \setminus S$ is the relation with

$$(a, b) \in R \setminus S \iff (a, b) \in R \text{ and } (a, b) \notin S.$$

Definition 0.4.19 (Complement of a Relation)**.** Let $R \subseteq A \times B$. The *complement* of R relative to $A \times B$ is

$$(A \times B) \setminus R.$$

Theorem 0.4.20 (Converse Laws for Relation Operations)**.** *Let $R, S \subseteq A \times B$, let $T \subseteq B \times C$, and take complements relative to the indicated Cartesian products. Then*

$$(R^{-1})^{-1} = R, \qquad (T \circ R)^{-1} = R^{-1} \circ T^{-1},$$
$$(R \cup S)^{-1} = R^{-1} \cup S^{-1}, \qquad (R \cap S)^{-1} = R^{-1} \cap S^{-1},$$
$$(R \setminus S)^{-1} = R^{-1} \setminus S^{-1}, \qquad ((A \times B) \setminus R)^{-1} = (B \times A) \setminus R^{-1}.$$

Go to proof.

Theorem 0.4.21 (Relation Composition and Boolean Operations)**.** *Let $P, Q \subseteq A \times B$, $R \subseteq B \times C$, and $S, T \subseteq B \times C$. Then*

$$R \circ (P \cup Q) = (R \circ P) \cup (R \circ Q), \qquad (S \cup T) \circ P = (S \circ P) \cup (T \circ P).$$

Also,

$$R \circ (P \cap Q) \subseteq (R \circ P) \cap (R \circ Q), \qquad (S \cap T) \circ P \subseteq (S \circ P) \cap (T \circ P).$$

The analogous formulas for difference and complement hold only as inclusions or under extra hypotheses; they are not general distributive laws. *Go to proof.*

Definition 0.4.22 (Transitive Closure)**.** Let R be a relation on A. The *transitive closure* of R is the smallest transitive relation on A containing R.

Definition 0.4.23 (Equivalence Closure)**.** Let R be a relation on A. The *equivalence closure* of R is the smallest equivalence relation on A containing R.

Definition 0.4.24 (One-to-Many Relation)**.** Let $R \subseteq A \times B$. R is *one-to-many* if some input is related to two distinct outputs:

$$\exists a \in A\, \exists b_1, b_2 \in B, \quad (a, b_1) \in R \wedge (a, b_2) \in R \wedge b_1 \neq b_2.$$

Definition 0.4.25 (Many-to-One Relation)**.** Let $R \subseteq A \times B$. R is *many-to-one* if two distinct inputs are related to a common output:

$$\exists a_1, a_2 \in A\, \exists b \in B, \quad (a_1, b) \in R \wedge (a_2, b) \in R \wedge a_1 \neq a_2.$$

Definition 0.4.26 (Many-to-Many Relation)**.** Let $R \subseteq A \times B$. R is *many-to-many* if it is one-to-many and many-to-one.

Definition 0.4.27 (Many-Place Relation)**.** A *ternary relation* among sets A, B, and C is a subset of $A \times B \times C$. More generally, an n-place relation on $A_1, \dots, A_n$ is a subset of

$$A_1 \times \cdots \times A_n.$$

For example, betweenness in geometry is naturally a three-place relation, and the arithmetic statement $x + y = z$ is naturally a three-place relation among x, y, and z.

0.4.2 Properties of Relations

Definition 0.4.28 (Reflexive Relation)**.** A binary relation $R \subseteq A \times A$ is *reflexive* on A if every element of the domain is related to itself:

$$\forall a \in A,\ (a, a) \in R.$$

In infix notation, this is the condition $\forall a \in A,\ aRa$.

Definition 0.4.29 (Irreflexive Relation)**.** A binary relation $R \subseteq A \times A$ is *irreflexive* on A if no element of the domain is related to itself:

$$\forall a \in A,\ (a, a) \notin R.$$

In infix notation, this is the condition $\forall a \in A,\ \neg(aRa)$.

Definition 0.4.30 (Symmetric Relation)**.** A binary relation $R \subseteq A \times A$ is *symmetric* on A if every related pair also appears in the reverse order:

$$\forall a, b \in A,\ (a, b) \in R \to (b, a) \in R.$$

In infix notation, this is $\forall a, b \in A,\ aRb \to bRa$.

Definition 0.4.31 (Antisymmetric Relation)**.** A binary relation $R \subseteq A \times A$ is *antisymmetric* on A if any two elements related in both directions must be equal:

$$\forall a, b \in A,\ \big((a, b) \in R \wedge (b, a) \in R\big) \to a = b.$$

In infix notation, this is $\forall a, b \in A,\ ((aRb \wedge bRa) \to a = b)$.

Definition 0.4.32 (Asymmetric Relation)**.** A binary relation $R \subseteq A \times A$ is *asymmetric* on A if every related pair excludes the reverse pair:

$$\forall a, b \in A,\ (a, b) \in R \to (b, a) \notin R.$$

In infix notation, this is $\forall a, b \in A,\ (aRb \to \neg(bRa))$.

Definition 0.4.33 (Transitive Relation)**.** A binary relation $R \subseteq A \times A$ is *transitive* on A if any two composable related pairs collapse to a direct related pair:

$$\forall a, b, c \in A,\ \big((a, b) \in R \wedge (b, c) \in R\big) \to (a, c) \in R.$$

In infix notation, this is $\forall a, b, c \in A,\ ((aRb \wedge bRc) \to aRc)$.

Definition 0.4.34 (Total (Connex) Relation)**.** R is *total* (or *connex*) if every pair is comparable:

$$\forall a, b \in A,\ (a, b) \in R \lor (b, a) \in R.$$

Definition 0.4.35 (Equivalence Relation)**.** R is an *equivalence relation* on A if it is reflexive, symmetric, and transitive.

Definition 0.4.36 (Preorder)**.** R is a *preorder* on A if it is reflexive and transitive.

Definition 0.4.37 (Partial Order)**.** R is a *partial order* on A if it is reflexive, antisymmetric, and transitive.

Definition 0.4.38 (Strict Partial Order)**.** R is a *strict partial order* on A if it is irreflexive and transitive.

Definition 0.4.39 (Total Order)**.** R is a *total order* on A if it is a partial order and is total.

0.5 Equivalence

0.5.1 Equivalence Classes and Partitions

Definition 0.5.1 (Equivalence Class)**.** Let R be an equivalence relation on A. For $a \in A$, the *equivalence class* of a is:

$$[a]_R \ :=\ \{\, x \in A \mid (a, x) \in R \,\}.$$

When R is clear from context, we write $[a]$.

Definition 0.5.2 (Quotient Set)**.** The *quotient set* of A by R is the set of all equivalence classes:

$$A/R \ :=\ \{\, [a]_R \mid a \in A \,\}.$$

Definition 0.5.3 (Index of an Equivalence Relation)**.** The *index* of R on A is the cardinality $|A/R|$, i.e. the number of equivalence classes.

Definition 0.5.4 (Canonical Surjection)**.** The *canonical surjection* (quotient map) is the function

$$\pi : A \to A/R, \qquad \pi(a) := [a].$$

Definition 0.5.5 (Partition)**.** A *partition* of A is a collection $\mathcal{P}$ of subsets of A such that:

(i) every block is nonempty: $\forall P \in \mathcal{P},\ P \neq \varnothing$;

(ii) distinct blocks are disjoint: $\forall P, Q \in \mathcal{P},\ P \neq Q \to P \cap Q = \varnothing$;

(iii) the blocks cover A: $\bigcup_{P \in \mathcal{P}} P = A$.

The sets $P \in \mathcal{P}$ are called the *blocks* of the partition.

Lemma 0.5.6 (Representative Independence Lemma)**.** *Let R be an equivalence relation on A. For any $a, b \in A$,*

$$[a] = [b] \iff (a, b) \in R.$$

Go to proof.

Theorem 0.5.7 (Equivalence Relations and Partitions)**.** *Let A be a set.*

(i) If R is an equivalence relation on A, then A/R is a partition of A.

(ii) If $\mathcal{P}$ is a partition of A, then the relation $R_{\mathcal{P}}$ defined by

$$(a, b) \in R_{\mathcal{P}} \iff \exists P \in \mathcal{P} \text{ with } a \in P \text{ and } b \in P$$

is an equivalence relation on A.

(iii) These constructions are inverse: $R_{A/R} = R$ *and* $A/R_{\mathcal{P}} = \mathcal{P}$.

Go to proof.

Theorem 0.5.8 (Universal Property of the Quotient Map)**.** *Let* R *be an equivalence relation on* A*, let* $\pi : A \to A/R$ *be the canonical surjection, and let* $f : A \to B$ *be any function. Then* f *is* constant on equivalence classes *— meaning* $(a, b) \in R$ *implies* $f(a) = f(b)$ *— if and only if there exists a unique function* $\bar{f} : A/R \to B$ *such that*

$$f = \bar{f} \circ \pi.$$

When it exists, $\bar{f}$ *is defined by* $\bar{f}([a]) = f(a)$. *Go to proof.*

Proofs

Functions

0.6 Functions — Quick Reference

0.6.1 Functions

Definition 0.6.1 (Functional Relation)**.** Let $R \subseteq A \times B$ be a relation. We say that R is *functional from A to B* if each input has at most one output:

$$\forall a \in A\, \forall b_1, b_2 \in B, \quad \big((a, b_1) \in R \wedge (a, b_2) \in R\big) \Rightarrow b_1 = b_2.$$

Definition 0.6.2 (Total Relation on a Domain)**.** Let $R \subseteq A \times B$ be a relation. We say that R is *total on A* if each input has at least one output:

$$\forall a \in A\, \exists b \in B, \quad (a, b) \in R.$$

Definition 0.6.3 (Function)**.** Let A and B be sets. A *function* from A to B is a relation $f \subseteq A \times B$, together with the declared domain A and codomain B, such that:

(i) (*Existence / left-total*) for every $a \in A$, there exists $b \in B$ with $(a, b) \in f$;

(ii) (*Uniqueness / right-unique*) if $(a, b_1) \in f$ and $(a, b_2) \in f$ then $b_1 = b_2$.

If $(a, b) \in f$ we write $f(a) = b$.

Theorem 0.6.4 (Functional Relation Characterization of Functions)**.** *Let A and B be sets, and let $R \subseteq A \times B$. Then R is a function from A to B if and only if*

$$\forall a \in A\, \exists! b \in B, \quad (a, b) \in R.$$

Equivalently, R is a function from A to B if and only if R is total on A and functional from A to B. *Go to proof.*

Definition 0.6.5 (Partial Function)**.** Let A and B be sets. A *partial function* from A to B is a functional relation $R \subseteq A \times B$. It need not be total on A: some inputs may have no output.

Proposition 0.6.6 (Partial Functions Are Functional Relations)**.** *For sets A and B and a relation $R \subseteq A \times B$, R is a partial function from A to B if and only if R is functional from A to B.* *Go to proof.*

Definition 0.6.7 (Function Equality)**.** For functions $f : A \to B$ and $g : C \to D$, we write $f = g$ if

$$A = C, \qquad B = D, \qquad \text{and} \qquad \forall a \in A,\ f(a) = g(a).$$

Proposition 0.6.8 (One-to-Many Relations Are Not Functional)**.** *If $R \subseteq A \times B$ is one-to-many, then R is not functional from A to B. Consequently, R is not a function from A to B.* *Go to proof.*

Definition 0.6.9 (Domain of a Function)**.** If f is a function from A to B, we write $f : A \to B$, where A is the *domain* $\operatorname{dom}(f)$.

Definition 0.6.10 (Codomain of a Function)**.** If f is a function from A to B, we write $f : A \to B$, where B is the *codomain* $\mathrm{cod}(f)$.

Definition 0.6.11 (Image of a Function)**.** The *image* (or *range*) of $f : A \to B$ is:

$$\mathrm{im}(f) \;:=\; \{\, b \in B \mid \exists a \in A,\ f(a) = b \,\}.$$

The image may be a proper subset of the codomain.

Definition 0.6.12 (Image of a Set)**.** For $S \subseteq A$, the *image of S under f* is:

$$f(S) \;:=\; \{\, f(a) \mid a \in S \,\}.$$

Note $f(A) = \mathrm{im}(f)$.

Definition 0.6.13 (Preimage)**.** For $T \subseteq B$, the *preimage* (inverse image) of T under f is:

$$f^{-1}(T) \;:=\; \{\, a \in A \mid f(a) \in T \,\}.$$

Definition 0.6.14 (Fiber)**.** The *fiber* of f over $b \in B$ is the preimage of the singleton:

$$f^{-1}(\{b\}) \;=\; \{\, a \in A \mid f(a) = b \,\}.$$

Definition 0.6.15 (Graph of a Function)**.** The *graph* of $f : A \to B$ is:

$$\mathrm{Graph}(f) \;:=\; \{\, (a, b) \in A \times B \mid b = f(a) \,\}.$$

Definition 0.6.16 (Injective Function)**.** $f : A \to B$ is *injective* (one-to-one) if distinct elements of A have distinct images:

$$\forall a_1, a_2 \in A,\ f(a_1) = f(a_2) \implies a_1 = a_2.$$

Definition 0.6.17 (Empty Function)**.** For every set B, the *empty function* from $\varnothing$ to B is the empty relation

$$\varnothing : \varnothing \to B.$$

It is the unique function with domain $\varnothing$ and codomain B.

Definition 0.6.18 (Surjective Function)**.** $f : A \to B$ is *surjective* (onto) if every element of B is achieved:

$$\forall b \in B,\ \exists a \in A \text{ such that } f(a) = b.$$

Equivalently, $\mathrm{im}(f) = B$.

Definition 0.6.19 (Bijective Function)**.** $f : A \to B$ is *bijective* if it is both injective and surjective.

Proposition 0.6.20 (One-One Relations and Converse Functionality)**.** *Let $f : A \to B$ be a function, identified with its graph $G_f \subseteq A \times B$. Then f is injective if and only if the converse relation $G_f^{-1} \subseteq B \times A$ is functional from B to A.* *Go to proof.*

Proposition 0.6.21 (Bijections Have Unique Correspondence Both Ways)**.** *Let $f : A \to B$ be a function. Then f is bijective if and only if*

$$\forall a \in A\, \exists! b \in B, \quad f(a) = b,$$

and

$$\forall b \in B\, \exists! a \in A, \quad f(a) = b.$$

Go to proof.

Definition 0.6.22 (Many-Place Functional Relation)**.** A relation $R \subseteq A_1 \times \cdots \times A_n \times B$ is *functional in the last coordinate* if

$$\forall a_1 \in A_1 \cdots \forall a_n \in A_n\, \forall b_1, b_2 \in B,$$
$$\big((a_1, \ldots, a_n, b_1) \in R \wedge (a_1, \ldots, a_n, b_2) \in R\big) \Rightarrow b_1 = b_2.$$

When such a relation is also total in the first n coordinates, it determines a function of n variables, written $f(a_1, \ldots, a_n) = b$.

Definition 0.6.23 (Identity Function)**.** The *identity function* on A is $\mathrm{id}_A : A \to A$, $\mathrm{id}_A(a) = a$.

Definition 0.6.24 (Inclusion Map)**.** If $A \subseteq B$, the *inclusion map* is $\iota : A \hookrightarrow B$, $\iota(a) = a$.

Definition 0.6.25 (Constant Function)**.** $f : A \to B$ is *constant* if there exists $b_0 \in B$ with $f(a) = b_0$ for all $a \in A$.

0.6.2 Composition, Inverses, and Set-Image Laws

Definition 0.6.26 (Composition)**.** Let $f : A \to B$ and $g : B \to C$. The *composition* $g \circ f : A \to C$ is:

$$(g \circ f)(a) := g\big(f(a)\big) \quad \text{for all } a \in A.$$

Theorem 0.6.27 (Associativity of Composition)**.** *For $f : A \to B$, $g : B \to C$, $h : C \to D$:*

$$h \circ (g \circ f) = (h \circ g) \circ f.$$

Go to proof.

Theorem 0.6.28 (Identity and Composition)**.** *For any $f : A \to B$:*

$$f \circ \mathrm{id}_A = f \quad \textit{and} \quad \mathrm{id}_B \circ f = f.$$

Go to proof.

Theorem 0.6.29 (Injectivity and Surjectivity Under Composition)**.** *Let $f : A \to B$ and $g : B \to C$.*

(i) If f and g are injective, then $g \circ f$ is injective.

(ii) If f and g are surjective, then $g \circ f$ is surjective.

(iii) If $g \circ f$ is injective, then f is injective.

(iv) If $g \circ f$ is surjective, then g is surjective.

Go to proof.

Definition 0.6.30 (Inverse Function)**.** Let $f : A \to B$ be bijective. The *inverse function* is $f^{-1} : B \to A$ defined by: $f^{-1}(b) = a$ iff $f(a) = b$.

Theorem 0.6.31 (Characterization of Inverse Functions)**.** *Let $f : A \to B$ be bijective. Then*

$$f^{-1} \circ f = \mathrm{id}_A \quad \textit{and} \quad f \circ f^{-1} = \mathrm{id}_B\,.$$

Conversely, a function admits an inverse iff it is bijective. *Go to proof.*

Theorem 0.6.32 (Inverse of a Composition)**.** *Let $f : A \to B$ and $g : B \to C$ be bijective. Then $g \circ f$ is bijective and*

$$(g \circ f)^{-1} = f^{-1} \circ g^{-1}.$$

Go to proof.

Definition 0.6.33 (Left Inverse)**.** Let $f : A \to B$. A *left inverse* of f is a function $g : B \to A$ with $g \circ f = \mathrm{id}_A$.

Definition 0.6.34 (Right Inverse)**.** Let $f : A \to B$. A *right inverse* (or *section*) of f is a function $h : B \to A$ with $f \circ h = \mathrm{id}_B$.

Theorem 0.6.35 (One-Sided Inverses and Function Properties)**.** *Let $f : A \to B$.*

(i) f has a left inverse $\iff$ f is injective (and $A \neq \varnothing$ or $A = B = \varnothing$).

(ii) f has a right inverse $\iff$ f is surjective.

(iii) If f has both a left inverse g and a right inverse h, then $g = h$ and f is bijective.

Go to proof.

Definition 0.6.36 (Restriction)**.** Let $f : A \to B$ and $C \subseteq A$. The *restriction* of f to C is $f|_C : C \to B$, $f|_C(a) = f(a)$ for all $a \in C$.

Definition 0.6.37 (Extension)**.** Let $A \subseteq A'$, and let $f : A \to B$. A function $g : A' \to B$ is an *extension* of f if $g|_A = f$.

Theorem 0.6.38 (Preimages Preserve Set Operations)**.** *Let $f : A \to B$ and $S, T \subseteq B$. Then*

(i) $f^{-1}(S \cup T) = f^{-1}(S) \cup f^{-1}(T)$,

(ii) $f^{-1}(S \cap T) = f^{-1}(S) \cap f^{-1}(T)$,

(iii) $f^{-1}(S \setminus T) = f^{-1}(S) \setminus f^{-1}(T)$,

(iv) $f^{-1}(S^c) = (f^{-1}(S))^c$.

Go to proof.

Theorem 0.6.39 (Images and Set Operations)**.** *Let $f : A \to B$ and $S, T \subseteq A$. Then*

(i) $f(S \cup T) = f(S) \cup f(T)$,

(ii) $f(S \cap T) \subseteq f(S) \cap f(T)$,

(iii) $f(S \setminus T) \supseteq f(S) \setminus f(T)$.

Equality holds in (ii) and (iii) for all S, T iff f is injective. *Go to proof.*

Theorem 0.6.40 (Image–Preimage Adjunction)**.** *Let $f : A \to B$, let $S \subseteq A$, and let $T \subseteq B$. Then*

$$S \subseteq f^{-1}(f(S)), \qquad f(f^{-1}(T)) \subseteq T.$$

Moreover, $S = f^{-1}(f(S))$ for every $S \subseteq A$ iff f is injective, and $f(f^{-1}(T)) = T$ for every $T \subseteq B$ iff f is surjective. *Go to proof.*

Proofs

Orderings

0.7 Order

0.7.1 Ordered Sets

Definition 0.7.1 (Ordered Set)**.** An *ordered set* is a set equipped with an order relation. Unless explicitly qualified, "ordered set" means a partially ordered set, and its relation is written $\leq$.

Definition 0.7.2 (Partially Ordered Set)**.** A *partially ordered set* is a pair $(A, \leq)$ where A is a set and $\leq$ is a partial order on A (a relation that is reflexive, antisymmetric, and transitive).

Definition 0.7.3 (Order-Structure Synonyms)**.** The following short forms abbreviate their full names and carry no independent mathematical content; each resolves to the canonical structure it names:

- *poset* := partially ordered set.
- *toset*, *loset*, *linearly ordered set* := totally ordered set (total order and linear order coincide, so these name one structure).
- *woset* := well-ordered set.

Definition 0.7.4 (Strict Order)**.** Let $(A, \leq)$ be an ordered set. The *strict order* $<$ is defined by:

$$a < b \iff (a \leq b \wedge a \neq b).$$

Definition 0.7.5 (Comparable and Incomparable Elements)**.** In an ordered set $(A, \leq)$, elements $a, b \in A$ are *comparable* if $a \leq b$ or $b \leq a$; they are *incomparable* if neither holds.

Definition 0.7.6 (Totally Ordered Set)**.** A *totally ordered set* (or *linearly ordered set*) is a pair $(A, \leq)$ where $\leq$ is a total order on A — equivalently, a partially ordered set in which every pair of elements is comparable:

$$\forall a, b \in A,\ a \leq b \vee b \leq a.$$

Definition 0.7.7 (Upper and Lower Bounds)**.** Let $(A, \leq)$ be an ordered set and $S \subseteq A$.
An element $u \in A$ is an *upper bound* of S if $\forall s \in S,\ s \leq u$.
An element $\ell \in A$ is a *lower bound* of S if $\forall s \in S,\ \ell \leq s$.

Definition 0.7.8 (Minimal Element)**.** Let $S \subseteq A$ in an ordered set $(A, \leq)$. An element $m \in S$ is a *minimal element* of S if there is no $s \in S$ with $s < m$.

Definition 0.7.9 (Maximal Element)**.** Let $S \subseteq A$ in an ordered set $(A, \leq)$. An element $M \in S$ is a *maximal element* of S if there is no $s \in S$ with $M < s$.

Definition 0.7.10 (Least Element)**.** Let $S \subseteq A$ in an ordered set $(A, \leq)$. An element $\ell \in S$ is the *least element* of S if

$$\forall s \in S,\ \ell \leq s.$$

Definition 0.7.11 (Greatest Element)**.** Let $S \subseteq A$ in an ordered set $(A, \leq)$. An element $g \in S$ is the *greatest element* of S if

$$\forall s \in S,\ s \leq g.$$

Definition 0.7.12 (Order-Preserving Map)**.** Let $(M, \leq)$ and $(M', \leq')$ be partially ordered sets. A function $f : M \to M'$ is *order-preserving* (or *monotone*) if

$$\forall a, b \in M, \quad a \leq b \implies f(a) \leq' f(b).$$

Definition 0.7.13 (Order Isomorphism)**.** Let $(M, \leq)$ and $(M', \leq')$ be partially ordered sets. A function $f : M \to M'$ is an *order isomorphism* if it is bijective and

$$\forall a, b \in M, \quad a \leq b \iff f(a) \leq' f(b).$$

Definition 0.7.14 (Well-Ordered Set)**.** A totally ordered set $(A, \leq)$ is *well-ordered* if every nonempty subset $S \subseteq A$ has a least element:

$$\forall S \subseteq A,\ \big(S \neq \varnothing \ \Rightarrow\ \exists m \in S \text{ s.t. } m \leq s\ \forall s \in S\big).$$

Definition 0.7.15 (Chain and Antichain)**.** A subset $C \subseteq A$ of a poset $(A, \leq)$ is a *chain* if every pair of elements in C is comparable. It is an *antichain* if no two distinct elements of C are comparable.

Definition 0.7.16 (Initial Segment)**.** A subset $I \subseteq A$ is an *initial segment* of $(A, \leq)$ if

$$a \in I \text{ and } b \leq a \ \Rightarrow\ b \in I.$$

Definition 0.7.17 (Directed Set)**.** A preorder $(I, \leq)$ is *directed* (or a *directed set*) if every finite subset of I has an upper bound in I: for all $i, j \in I$ there exists $k \in I$ with $i \leq k$ and $j \leq k$.

Definition 0.7.18 (Net)**.** Let $(I, \leq)$ be a directed set and X a set. A *net* in X is a function $x : I \to X$. The value at $i \in I$ is written x_i, and the net is written $(x_i)_{i \in I}$.

0.7.2 Supremum and Infimum

Definition 0.7.19 (Supremum and Infimum)**.** Let $(A, \leq)$ be a poset and $S \subseteq A$.
The *supremum* (or *least upper bound*) of S, written $\sup S$, is an element $u^* \in A$ satisfying:

(i) u^* is an upper bound of S: $\forall s \in S,\ s \leq u^*$;

(ii) u^* is the *least* upper bound: if u is *any* upper bound of S, then $u^* \leq u$.

The *infimum* (or *greatest lower bound*) of S, written $\inf S$, is an element $\ell^* \in A$ satisfying:

(i) ℓ^* is a lower bound of S: $\forall s \in S,\ \ell^* \leq s$;

(ii) ℓ^* is the *greatest* lower bound: if ℓ is *any* lower bound of S, then $\ell \leq \ell^*$.

Proposition 0.7.20 (Uniqueness of Supremum and Infimum)**.** *Let $(A, \leq)$ be a poset and $S \subseteq A$. If $\sup S$ exists, it is unique. If $\inf S$ exists, it is unique.*

Proposition 0.7.21 (Characterisation of Supremum)**.** *Let $(A, \leq)$ be a poset, $S \subseteq A$, and $u^* \in A$. Then $u^* = \sup S$ if and only if:*

(i) u^* is an upper bound of S; and

(ii) for every $v \in A$ with $v < u^*$, there exists $s \in S$ with $v < s$.

Proposition 0.7.22 (Duality of Supremum and Infimum)**.** *Let $(A, \leq)$ be a poset and $S \subseteq A$. Then $\inf_{(A,\leq)} S = \sup_{(A,\geq)} S$, where $(A, \geq)$ denotes the dual poset.*
In particular: if every nonempty bounded-above subset of $(A, \leq)$ has a supremum, then every nonempty bounded-below subset of $(A, \leq)$ has an infimum.

0.7.3 Lattices

Definition 0.7.23 (Lattice)**.** A *lattice* is a poset $(L, \leq)$ in which every two-element subset $\{a, b\} \subseteq L$ has both a supremum and an infimum in L.

The supremum $\sup\{a, b\}$ is called the *join* of a and b, written $a \vee b$. The infimum $\inf\{a, b\}$ is called the *meet* of a and b, written $a \wedge b$.

Definition 0.7.24 (Complete Lattice)**.** A lattice $(L, \leq)$ is a *complete lattice* if every subset $S \subseteq L$ (including the empty set and L itself) has both a supremum and an infimum in L:

$$\forall S \subseteq L, \quad \sup S \in L \text{ and } \inf S \in L.$$

Definition 0.7.25 (Distributive Lattice)**.** A lattice $(L, \leq)$ is *distributive* if for all $a, b, c \in L$:

$$a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c).$$

Definition 0.7.26 (Complement and Boolean Lattice)**.** Let $(L, \leq)$ be a bounded distributive lattice with top element $\top$ and bottom element $\bot$. An element $a' \in L$ is a *complement* of $a \in L$ if

$$a \vee a' = \top \quad \text{and} \quad a \wedge a' = \bot.$$

A *Boolean lattice* (or *Boolean algebra*) is a bounded distributive lattice in which every element has a complement.

0.7.4 Hasse Diagrams and the Duality Principle

Definition 0.7.27 (Cover Relation)**.** Let $(A, \leq)$ be a poset and $a, b \in A$. We say b *covers* a, written $a \lessdot b$, if $a < b$ and there is no $c \in A$ with $a < c < b$.

Definition 0.7.28 (Hasse Diagram)**.** The *Hasse diagram* of a finite poset $(A, \leq)$ is the directed graph whose vertices are the elements of A, with an upward edge from a to b whenever $a \lessdot b$. Three graphical conventions apply:

(i) Reflexive loops ($a \leq a$) are suppressed.

(ii) Edges implied by transitivity are suppressed: if $a \lessdot b$ and $b \lessdot c$, no direct edge from a to c is drawn.

(iii) Direction is encoded by height: $a \lessdot b$ is drawn with b strictly above a, so arrowheads are unnecessary.

Definition 0.7.29 (Dual Poset)**.** Let $(A, \leq)$ be a poset. The *dual poset* is $(A, \geq)$, where $a \geq b$ is defined to mean $b \leq a$. The dual is also written $(A, \leq^{\mathrm{op}})$.

Proposition 0.7.30 (Duality Principle for Posets)**.** *Let Φ be any first-order statement about a poset $(A, \leq)$ expressed using only the relation $\leq$. Let Φ^* be the statement obtained from Φ by replacing every occurrence of $\leq$ with $\geq$ (equivalently, working in the dual poset). Then:*

$$(A, \leq) \models \Phi \iff (A, \geq) \models \Phi.$$

In particular, if Φ is a theorem about all posets, then so is Φ^*.

0.7.5 Order Extensions

Definition 0.7.31 (Symmetric and Asymmetric Parts)**.** For any binary relation R on X, define:

- the *symmetric part*: $xIy \iff xRy$ and yRx
- the *asymmetric part*: $xPy \iff xRy$ but not yRx

Note that $R = P \cup I$.

Definition 0.7.32 (Maximal Elements and Upper Bounds)**.** Let $\succsim$ be a binary relation on X.

- The set of *maximal elements* of X is

$$\mathrm{Max}(X, \succsim) = \{x \in X \mid y \succsim x \text{ for no } y \in X \text{ with } y \neq x\}.$$

- The set of *upper bounds* of X is

$$M(X, \succsim) = \{x \in X \mid x \succsim y \text{ for all } y \in X\}.$$

Definition 0.7.33 (Extension of a Preorder)**.** Let $\succsim$ be a preorder on X. A preorder $\succsim'$ is an *extension* of $\succsim$ if

$$x \succsim y \;\Rightarrow\; x \succsim' y, \qquad x \succ y \;\Rightarrow\; x \succ' y.$$

A *complete extension* is an extension that is also complete.

Corollary 0.7.34 (Complete preorder extension)**.** *For any nonempty set X and preorder $\succsim$ on X, there exists a complete preorder that is an extension of $\succsim$.*

Definition 0.7.35 (and Proposition (Only Weak Cycles))**.** Let $\succsim$ be a binary relation on X with asymmetric part $\succ$ and transitive closure $T(\succsim)$. We say $\succsim$ satisfies *only weak cycles* (OWC) if

$$xT(\succsim)y \;\Rightarrow\; \neg(y \succ x).$$

Proposition. A binary relation $\succsim$ on a nonempty set X can be extended to a complete preorder if and only if it satisfies OWC.

0.7.6 Induced Orders and Order Embeddings

Definition 0.7.36 (Induced Order)**.** Let $(B, \leq')$ be a partially ordered set and let $f : A \to B$ be a function. The *order induced by f* (or *pullback order along f*) is the relation $\leq_f$ on A defined by:

$$x \;\leq_f\; y \iff f(x) \;\leq'\; f(y).$$

Proposition 0.7.37 ($\leq_f$ is always a preorder)**.** *For any function $f : A \to B$ and partial order $(B, \leq')$, the relation $\leq_f$ is a preorder on A: it is reflexive and transitive.*

Definition 0.7.38 (f-Indistinguishable Elements)**.** Let $f : A \to B$ and $\leq_f$ be the induced order. Two elements $x, y \in A$ are *f-indistinguishable*, written $x \sim_f y$, if both $x \leq_f y$ and $y \leq_f x$, i.e. if

$$f(x) \leq' f(y) \quad \text{and} \quad f(y) \leq' f(x).$$

Proposition 0.7.39 ($\leq_f$ is a partial order iff f is injective)**.** *Let $(B, \leq')$ be a partially ordered set and $f : A \to B$. The induced order $\leq_f$ is a partial order on A if and only if f is injective.*

Definition 0.7.40 (Suborder / Restriction)**.** Let $(B, \leq')$ be a partially ordered set and $S \subseteq B$. The *suborder* on S (or *induced order on S*) is the relation $\leq'_S$ defined by:

$$x \;\leq'_S\; y \iff x \leq' y, \quad \text{for } x, y \in S.$$

Definition 0.7.41 (Order Embedding)**.** Let $(A, \leq)$ and $(B, \leq')$ be partially ordered sets. A function $f : A \to B$ is an *order embedding* if for all $x, y \in A$:

$$x \;\leq\; y \iff f(x) \;\leq'\; f(y).$$

Proposition 0.7.42 (Order embeddings are injective)**.** *Every order embedding $f : (A, \leq) \to (B, \leq')$ is injective.* *Go to proof.*

Proposition 0.7.43 (Order embedding is isomorphism onto image)**.** *If $f : (A, \leq) \to (B, \leq')$ is an order embedding, then f is an order isomorphism from $(A, \leq)$ to the suborder $(f(A), \leq'_{f(A)})$.* *Go to proof.*

0.8 Zorn

0.8.1 Zorn's Lemma and the Axiom of Choice

Proofs

Cardinality

0.9 Cardinality

0.9.1 Cardinality and Infinite Sets

Definition 0.9.1 (Same Cardinality)**.** Two sets A and B have the *same cardinality*, written $A \sim B$, if there exists a bijection $f : A \to B$.

Definition 0.9.2 (Finite and Infinite Sets)**.** A set A is *finite* if $A = \varnothing$ or $A \sim \{1, 2, \dots, n\}$ for some $n \in \mathbb{N}$. A set is *infinite* if it is not finite.

Definition 0.9.3 (Countable and Uncountable Sets)**.** A set A is *countably infinite* if $A \sim \mathbb{N}$. A set is *countable* if it is finite or countably infinite. A set is *uncountable* if it is infinite but not countable.

Definition 0.9.4 (Comparing Cardinalities)**.** Write $|A| \leq |B|$ if there exists an injection $f : A \hookrightarrow B$. Write $|A| < |B|$ if $|A| \leq |B|$ but $A \not\sim B$.

Theorem 0.9.5 ($\mathbb{Q}$ is countable)**.** *The set $\mathbb{Q}$ of rational numbers is countable.* *Go to proof.*

Theorem 0.9.6 (Countable union of countable sets is countable)**.** *Let $\{A_n\}_{n\in\mathbb{N}}$ be a countable family of countable sets. Then $\bigcup_{n=1}^{\infty} A_n$ is countable.* *Go to proof.*

Theorem 0.9.7 ($\mathbb{R}$ is uncountable)**.** *The set $\mathbb{R}$ of real numbers is not countable.* *Go to proof.*

Theorem 0.9.8 (Cantor's Theorem)**.** *For any set A, there is no surjection $A \to \mathcal{P}(A)$. In particular, $|A| < |\mathcal{P}(A)|$.* *Go to proof.*

Definition 0.9.9 (Function Space B^A)**.** For sets A and B, the *function space* B^A denotes the set of all functions from A to B:

$$B^A \;:=\; \{\, f : A \to B \,\}.$$

Theorem 0.9.10 (Schröder–Bernstein Theorem)**.** *If $|A| \leq |B|$ and $|B| \leq |A|$, then $A \sim B$.*
Equivalently: if there exist injections $f : A \hookrightarrow B$ and $g : B \hookrightarrow A$, then there exists a bijection $h : A \to B$. *Go to proof.*

Proofs